

The application of Graphene on Energy Storage Systems

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Abstract:

With the increasing demand for improved performance energy storage devices and the shift toward greener energy resources have laid the foundation for research interest on developing advance energy storage devices and conversion system. Since the discovery of graphene in 2004 which is a two-dimensional allotrope of carbon, have become a center of attraction among scientist due to its high specific surface area, good chemical stability and electrical properties. It has been considered as one of the biggest thing of twenty first century. This Review focuses on the application of graphene and graphene-based materials for energy storage systems like lithium-ion batteries, Sodium-ion batteries, supercapacitors and fuel cells.

Organization	22 (-)
Introduction	22 (-)
Expt Setup / Analysis	22 (+)
Conclusion	20 (-)
Total	86 (82)

1. Introduction

The most threatening problem our world is facing today is climate change. The adverse effect of nature is threatening our existence. The ecological balance is now on the verge of destruction because of the natural calamity's occurrence throughout the world and it is increasing day by day. The main reason behind this is the burning of fossil fuel to power the society. The world is now more diverse and huge advancement throughout the world results in the surge of power demand. To meet this requirement countries are burning more fossil fuel committing to more pollution and creating more greenhouse gas. To solve this problem, we need to reduce the consumption of fossil fuel and shift to a sustainable one. Following the 2nd world war and later nuclear power has been the forefront of the sustainable energy and most first world countries energy production are vastly depending on this. Though it was thought of safe and reliable technology, several incidents like Chernobyl and Fukushima have made scientist to think again and eventually look for other alternative technology. The only way to overcome this challenge is to develop renewable energy technology. In order to make the effective use of renewable energy, it is important to develop high-performance, low-cost and environmental-friendly energy conversion and storage systems. Energy storage is a vital issue to be addressed within this scenario, and electrochemical energy storage systems are undoubtedly a key enabler. Lithium-ion batteries (LIBs), supercapacitors (SCs), and other rechargeable batteries are the most prominent and fast-growing energy storage devices in view of their high energy and power density, long cycle life, and environmental aspect. A common feature of these energy storage devices is that they all consist of an electrolyte with separator and two electrodes where electron and ion conduction are highly required for involved electrochemical active materials¹. Although the energy storage mechanisms are different, design nanostructured electrode materials and electrode systems with excellent electrochemical properties has proven to be critical for tremendous advancements in these energy storage technologies. Nanostructured electrode materials hold the advantages of large electrode electrolyte contact area, short transport lengths for electrons, which are not possible with their bulk counterparts. Graphene and graphene based materials have received considerable attention and have been widely investigated for this various energy storage systems.

There are many available carbon allotropes, such as buckyminsterfullerene, carbon nanotube and graphene. Among them, graphene, a single layer of two-dimensional honeycomb carbon lattice, is one of the carbon family is emerged as exciting novel material. Graphene has unique properties, including high specific surface area, good chemical stability and excellent electrical conductivity². These properties make graphene to be an excellent candidate for energy conversion and storage applications. Here, we will focus on graphene as a material for energy conversion and storage applications with a brief overview on graphene and then the electrochemical applications of graphene in energy conversion and storage devices, including fuel cells and supercapacitors.

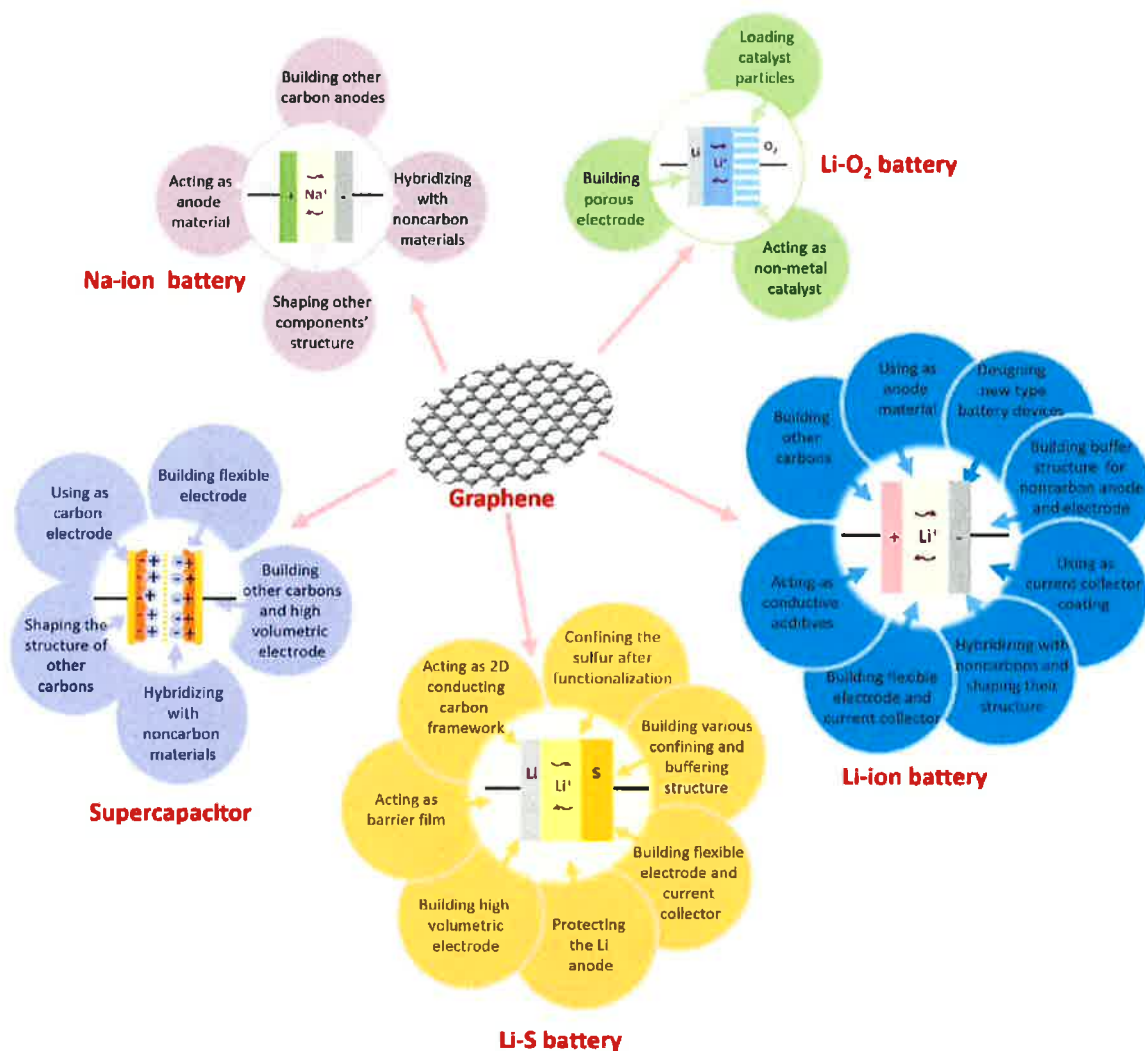


Figure 1. The applications of graphene in different Energy Storage Devices¹.

2. Graphene

Graphene is a two-dimensional one-atom-thick planar sheet of sp^2 bonded carbon atoms³, which is considered as the fundamental foundation for all fullerene allotropic dimensionalities, as shown in Fig. 2. In addition to its planar state graphene can be ‘wrapped’ into zero-dimensional spherical buckyballs, ‘rolled’ into one-dimensional CNTs – further categorized into single- or multi- walled depending on the number of graphene layers present (SWCNTs/MWCNTs respectively), or can be ‘stacked’ into three-dimensional graphite – generally consisting of more than ten graphene layers⁴. Consequently graphene can be considered the ‘mother of all carbon forms’ – as a building block. A variation on graphene are nano-platelets which are characterized by stacks consisting of between two and ten graphene sheets, with another graphene existing in the form of graphene oxide (GO) – where the graphene has been oxidized within the employed fabrication process or spontaneously by contact with air, however, this form is usually chemically or electrochemically

reduced before use. It is important to clarify that a single graphene nanosheet (GNS) refers to graphene as its standard form – that is a single layer of graphene.

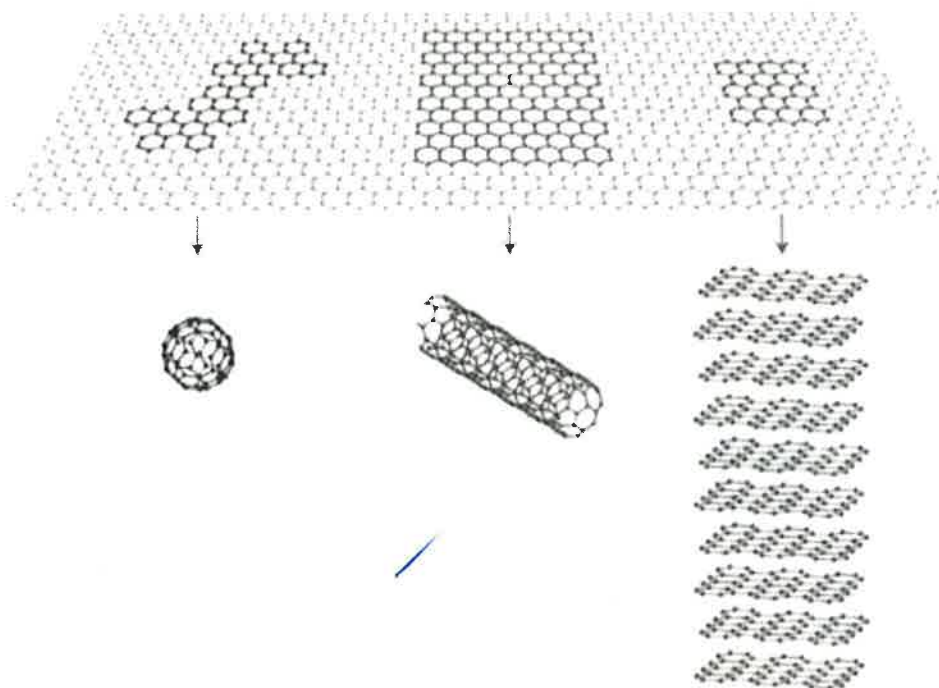


Figure 2. Schematic of Graphene³.

Since graphene was discovered by mechanical exfoliation using Scotch-tape method in 2004, many ways have been developed to achieve a high-quality and large-area graphene. Information about how the graphene was prepared is relevant, because the properties of graphene strongly depend on preparation process. The reported methods are generally fit into two major approaches that are “top-down” and “bottom-up” approaches. The meaning of top-down approach implies the exfoliation of natural or synthetic graphite into the mixture of a single and a few layer graphene platelets. Graphite into graphite oxide, which is precursor of graphene oxide (GO), is the most representative example of the top-down approach. Also, liquid-phase exfoliation, graphite intercalation compounds (GICs) and electrochemical exfoliation are all included in top-down approaches. These approaches have the advantages for scalable and low-cost production. However, it is hard to obtain single-layer graphene with high-quality because of defect creation during exfoliation like dangling bond, stone wales defect. In particular, reduction of GO into graphene results in a graphitic structure that is also one-atom thick, but it still contains large number of defects, such as nanoholes and Stone–Wales defect (heptagon/pentagon bonded carbon atom network). On the other hand, bottom-up approaches, which are directly growing graphene from organic precursors such as methane and other hydrocarbon sources, include epitaxial growth⁵ and chemical vapor deposition (CVD)⁶. Bottom-up approach is one of the most attractive methods for high-quality and large-area graphene production. Particularly, CVD grown graphene has been demonstrated to be of promise for potential applications as transparent electrodes. However, bottom-up approaches are extremely difficult to control manufacturing process, and thus they result in high manufacturing cost.

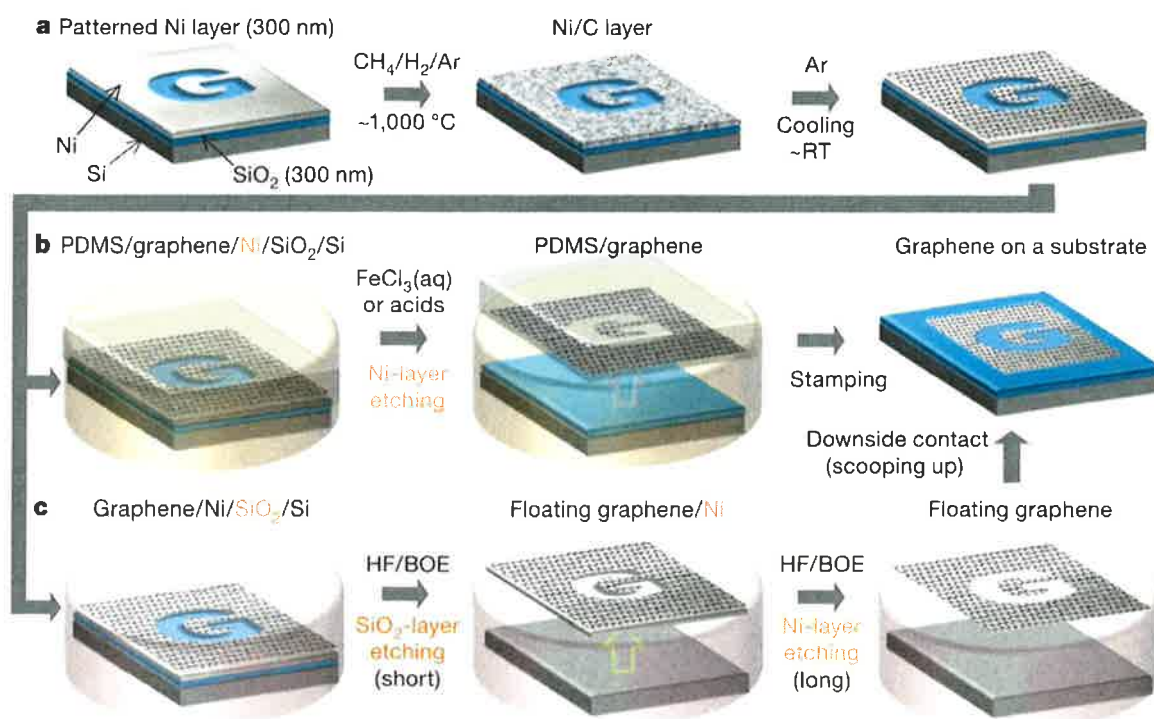


Figure 3. (a) Synthesis, (b, c) Etching and Transfer processes for graphene films⁶.

For energy storage applications, the device performance strongly depends on the properties of the materials. The top-down approach normally includes the methods of mechanical exfoliation by scotch tape and solution-based exfoliation, chemical exfoliation and reduction, and so on. Because of the low yield, mechanical exfoliation by scotch tape method is unsuitable for energy storage applications although the graphene prepared has the best quality with higher charge carrier mobility. The bottom-up approach usually has the same problem. The advantages and disadvantages of different preparation methods can be seen in Fig. 4 but ways to obtain a high yield must be found for practical applications.

Graphene has a potential for commercialization of the various devices due to its outstanding properties. However, it has an inherent limitation for pristine single-layer graphene (mechanically exfoliated graphene) to fabricate devices, specifically electrochemical devices. The properties of graphene from mechanical exfoliation are the best compared to the GO and other synthetic graphene materials. But, most of the literatures report energy devices on the basis of GO and doped-graphene, because of their availability at low-cost and scalability. Energy conversion and storage devices are required for large-scale production. GO is usually manufactured by breaking apart graphite into the sheets of oxidized graphene *via* acid intercalation, oxidation and subsequent physical agitation typically by sonication to yield GO nanosheets⁶. This approach is relatively simple and easy for mass production compared with CVD. In addition, graphite, which is starting material for GO, is sold at a very low price at about a few cents per gram. So, CVD is the most applicable method of production of graphene and graphene materials.

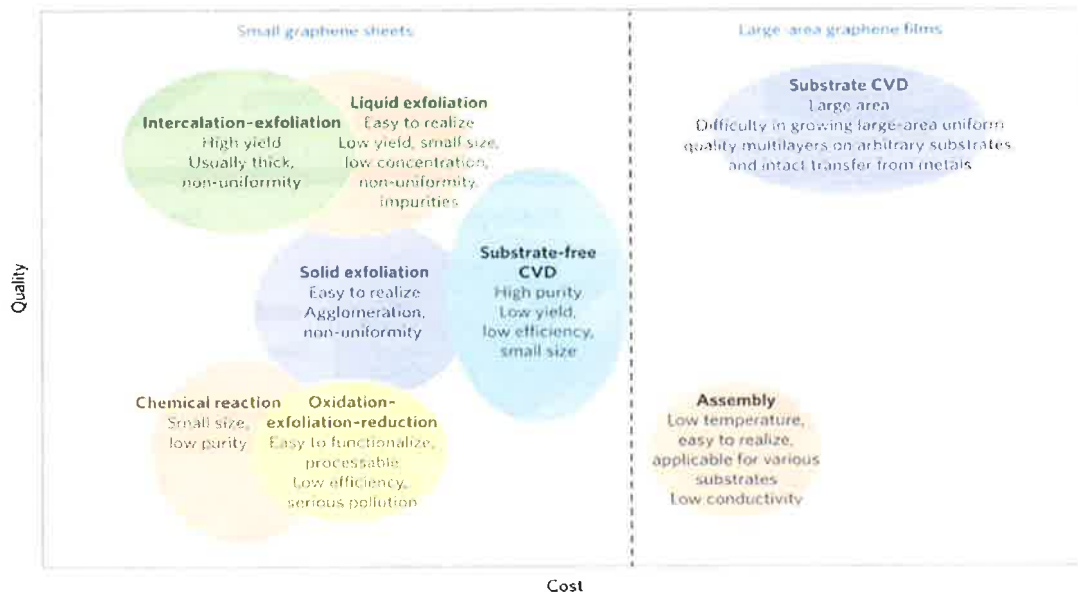


Figure 4. Cost Comparison of Graphene with Quality⁷.

3. Application of Graphene on Energy Storage Devices:

3.1 Batteries:

The lithium-ion battery (LIB), as an effective electrochemical energy storage device, has attracted great interests due to their high energy density, longer life and better safety compared to the traditional batteries like lead acid battery that have been developed over the years. However, although their energy density is high, LIBs still suffer from a much lower power density compared to electrochemical capacitors (ECs), another important energy storage device, and minimum of 6 hr charging time is normally necessary for the mobiles phones or laptops that are equipped with a lithium ion battery in the present market. To simultaneously achieve high power and large energy density at fast charge and discharge rates from several minutes to seconds still remains a great science and engineering challenge for the development of LIB. The power capability of a LIB depends critically on the speed at which Li ions and electrons migrate through the electrolyte and electrode. Therefore, the key to create high-power LIBs is developing new materials with high electrical conductivity for fast electron transport and a large surface area and well-developed nanostructures with shortened diffusion length for Li ions⁸. In this aspect, graphene, due to its superior electrical conductivity, excellent mechanical flexibility, good chemical stability, and high surface area, is expected to be a good candidate. Graphene based cathode and anode are applied to most the rechargeable battery today like LIB, Sodium-ion based battery, Magnesium-ion based battery, Zinc based battery. The main problem is to ensure the smooth intercalation of ions in anode and cathode during charging and discharging. An example like for Sodium ion battery the size of the Sodium-ion is larger than the Lithium, so this requires more intercalation space between graphene sheets. So the electrode used for LIB cannot be applied here which demands some special type of electrodes made of graphene. For anodes there are several different types of materials have been developed. A phosphorene-graphene hybrid material as a high-capacity anode for sodium-

ion batteries have been developed and showing promising future for Sodium batteries⁹. Because Lithium reserve is less in nature and as the electric vehicle (EV) and energy storage are evolving the demand for lithium is huge and it might pose a threat in near future as the reserve gets depleted. The study shows the phosphorene nanosheets, with increased interlayer distance, offer a short and effective diffusion distance for sodium ions. Moreover, the graphene enhances both the electrical

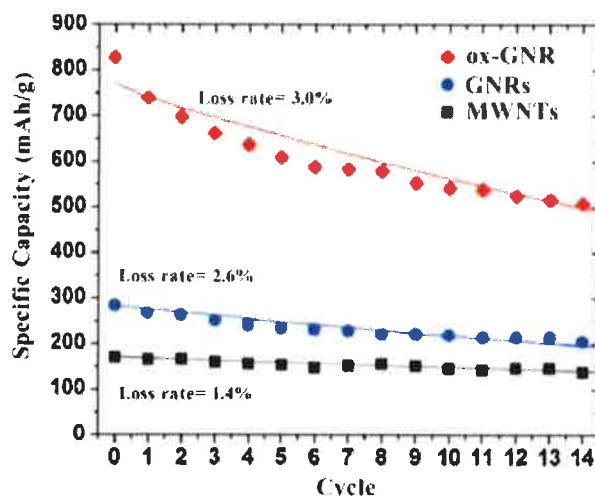


Figure 5. Cyclability of three different carbonaceous electrodes materials; oxidized graphene (ox-GNR); graphene (GNR); and multi-walled carbon nanotubes (MWCNTs)¹⁰.

conductivity of the material and provides a preferential pathway to the electrons generated by the redox reaction of phosphorene. The electrochemical properties of the sandwiched phosphorene-graphene hybrid could render it a suitable anode material for sodium-ion batteries. Another research shows the expanded graphite as superior anode for sodium-ion batteries make the battery more useful¹¹. So Sodium ion batteries can play an important role on that and this electrodes are showing promising future for that.

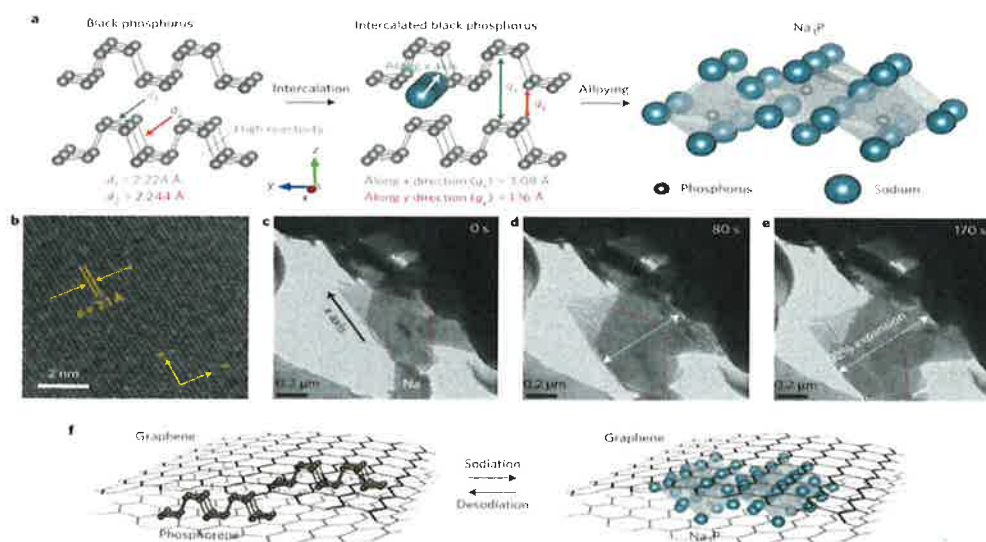


Figure 6. Sodiation in Phosphorene-Graphene Hybrid Material⁹.

3.2 Supercapacitors

Supercapacitors are a valuable commodity in the field of electronics and can store up more than a hundred times more energy than standard capacitors. Supercapacitors are also able to work in low temperature conditions and are regularly used in lieu of electrochemical batteries. One of the main properties of a supercapacitance material is the ability to form double-electric layers. This is essential in electric double-layer capacitors (EDLC) supercapacitors. Supercapacitors work by accumulating charges at the electrode-electrolyte interface through polarization, so that the energy is stored. Activated carbon has been the traditional electrode choice, but it suffers from the inability to operate at high voltages. Graphene, and its derivatives, are useful due to their high specific surface area, high conductivity, open-pore structure, production potential and low cost; all of which are desirable properties for a supercapacitor¹².

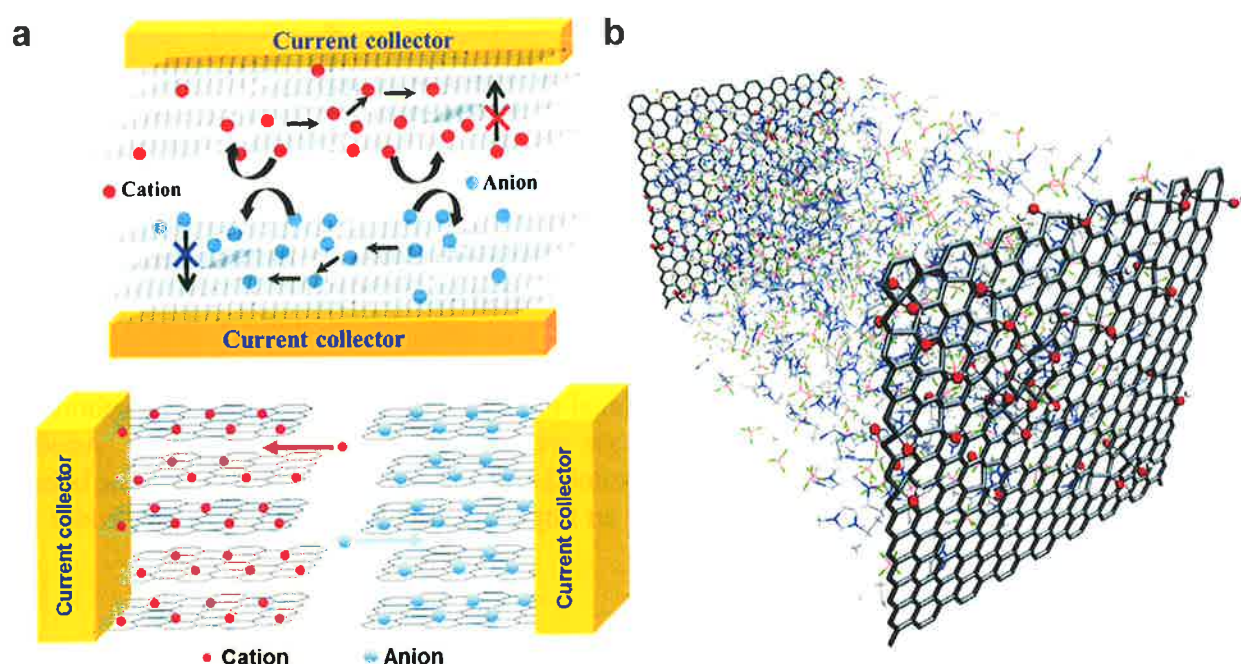


Figure 7. (a) Conventional stacked geometry used for carbon based electrode material (top) and in-plane supercapacitor configuration made of graphene electrodes (bottom). (b) Representation of a supercapacitor model consisting of GO electrodes¹³.

3.3 Fuel Cells

In addition to the tremendous impact that graphene has in the field of energy storage, graphene has also made a significant impact in the fabrication and application of energy generation devices. With the World's climate on the conscience of many and the depletion of non-renewable energy sources ever-nearing, the search for replacements has resulted in great interest in fuel cell applications and consequentially the examination and development of new electrode materials.

The fuel cell technology can be employed in many applications and has the tendency to become a vital part of the future energy system. Fuel cell has drawn great attention as an upcoming alternative source of energy. The process of fuel cells involves the production of water, electricity, and heat without any toxic or pollutant byproduct. Graphene is attractive as an electrode material and separation membrane for use within fuel cells because of its large surface area, improved enzymatic binding ability, unique electrical conductivity, widely applicable electro-catalytic activity, and low production costs. Graphene can be used for fuel cell as catalyst support or metal free catalyst and ionomer after proper chemical modification¹⁴.

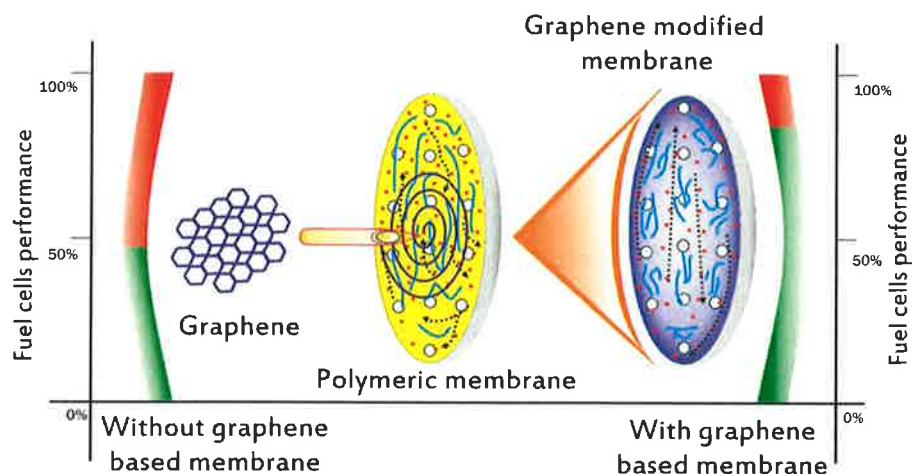


Figure 8. Performance of Graphene in Fuel Cells¹⁵.

4. Conclusion:

From the beginning of the graphene era, the push to find new applications for graphene material is more vibrant than ever. The past decades have witnessed the rapid development of numerous technologies in the field of energy conversion and storage. The application of graphene and its derivatives into the traditional active materials have brought about many remarkable advances in the energy storage systems. Graphene promises to increase substantially the energy and power density of practical systems, as well as enable the development of next-generation devices.

5. References:

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